

ICU Mortality Prediction Using the eICU Collaborative Research Database

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1 Introduction

Intensive Care Units (ICUs) manage patients with severe and life-threatening conditions who require continuous monitoring and complex medical interventions. Early identification of patients at high risk of death is a crucial challenge in critical care medicine, as it may support clinical decision-making, improve resource allocation and facilitate timely interventions. Reliable mortality prediction models can therefore provide valuable support to healthcare professionals by helping prioritize patients who may benefit from closer monitoring or more aggressive treatment strategies.

The objective of this project was to develop and evaluate machine learning models capable of predicting in-hospital mortality using only data collected during the first 24 hours of ICU admission. The analysis was performed using the eICU Collaborative Research Database, a large multicenter critical care dataset containing information from hospitals across the United States during 2014 and 2015.

The problem was formulated as a binary classification task, where patients were classified according to their survival status at ICU and hospital discharge.

2 Materials and Methods

2.1 Dataset Overview

The analysis was based on 8 source tables including patient demographics, diagnoses, laboratory measurements, intake and output records and physiological monitoring data. Each table contained patient records identified by different codes, providing a comprehensive clinical overview of each subject. Since these datasets originated from different hospitalizations and some clinical signs were recorded over time, they were longitudinal in nature, containing multiple rows per patient. A careful selection and aggregation procedure was therefore required before proceeding.

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2.2 Feature Extraction and Engineering

The first step, applied to all datasets, was removing all information collected after the first 24 hours of ICU stay. Each table was then analyzed and preprocessed individually:

- **Patient dataset:** contained 2,520 ICU records derived from different hospitalizations. Stays shorter than 24 hours were removed, and a single admission per patient was retained, giving priority to admissions associated with mortality outcomes and, when multiple eligible admissions were available, to the most recent hospitalization. This approach ensured patient independence and avoided information leakage. The total number of ICU admissions per patient was preserved as an additional feature (**ICUcounts**), capturing care complexity that would otherwise be lost. This dataset also revealed a strong class imbalance in the target variable, with only approximately 10% of patients classified as deceased.
- **Diagnosis dataset:** diagnosis information was extracted from ICD-9 codes and grouped into macro-categories using the first three digits of the code, then encoded via priority-weighted one-hot encoding (**Primary** = 3, **Major** = 2, **Other** = 1, **None** = 0). Major disease groups included circulatory disorders, respiratory diseases, endocrine conditions, neoplasms and infectious diseases.
- **Laboratory dataset:** identical tests were sometimes reported using different units of measurement across hospitals. After standardizing units, laboratory variables were aggregated at patient level using mean, minimum and maximum values observed during the first 24 hours. Minimum and maximum values were preferred over standard deviation as they better capture the range of clinical measurements.
- **Intake and output datasets:** total fluid intake, fluid output, dialysis-related measurements and body weight were aggregated at patient level.
- **Vital signs datasets (periodic and aperiodic):** variables such as heart rate, respiratory rate, oxygen saturation, systolic blood pressure, diastolic blood pressure and mean arterial pressure were summarized using mean, minimum and maximum values.

Additional clinically meaningful variables, such as pulse pressure ($PP = \text{non_invasive_systolic} - \text{non_invasive_diastolic}$), were derived from the original measurements.

- **Infusion drug dataset:** contained drug administration records for only 451 patients, with dosage expressed in different units depending on each patient’s weight and clinical needs. Given that this information was unavailable for more than half of the cohort, this dataset was excluded from the feature matrix to avoid introducing excessive missingness.
- **Hospital dataset:** contained logistic and structural information about each hospital, such as number of beds and teaching status. This dataset was reserved for the stratified performance analysis and was not merged into the feature matrix.

Features extracted from all retained tables were merged at patient level using an inner join strategy, retaining only patients present across all tables. Body Mass Index (BMI) was additionally derived from height and weight measurements to replace those two columns with a more clinically informative composite indicator. The target variable was constructed using both ICU discharge status and hospital discharge status; in case of inconsistencies, ICU discharge information was considered the most reliable source. The full cohort selection procedure, from the initial 2,520 ICU records to the final cohort of 1,003 patients, is summarized in Figure 1.

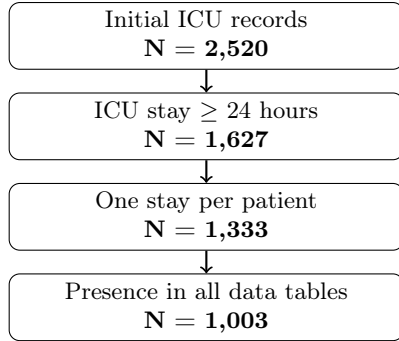


Fig. 1: Cohort selection flowchart.

2.3 Data Preprocessing

The merged dataset finally obtained contained 1003 patients and 80 features columns. It underwent an additional preprocessing before model development. Duplicate observations were removed, physiologically implausible values were corrected (for instance, this happened to impossible heart rate values or unrealistic oxygen saturation measurements) and variables with extremely high percentages of missing data were excluded from the analysis.

Exploratory Data Analysis (EDA) was performed to investigate the structure of the dataset. Numerical variables were analyzed through distribution plots and Pearson cor-

relation matrices, while categorical variables were evaluated using Cramér’s V association measures. Relationships between numerical and categorical variables were quantified using ANOVA eta-squared statistics, while outlier detection was performed using the interquartile range method.

The correlation analysis revealed several strongly correlated laboratory variables. To reduce multicollinearity and improve model stability, redundant features were removed while preserving clinically interpretable variables.

The final dataset was divided into training (60%), validation (20%) and test (20%) sets using stratified sampling to preserve the original mortality distribution across all partitions.

To prevent data leakage, an automated preprocessing pipeline was constructed and fitted exclusively on the training set, then applied to the validation and test sets. The pipeline included Multiple Imputation by Chained Equations (MICE) to estimate missing values while preserving relationships among variables, followed by z-score normalization of numerical features. For categorical features, the imputation was instead performed using their mode.

Feature selection was subsequently performed using the Minimum Redundancy Maximum Relevance (mRMR) algorithm, which selects variables that are strongly associated with the outcome while minimizing redundancy among predictors.

2.4 Model Development

Four supervised machine learning algorithms were evaluated:

- Lasso Logistic Regression
- Random Forest
- Support Vector Machine
- XGBoost

Hyperparameter optimization was conducted through grid search combined with 5-fold stratified cross-validation on the training set, maximizing PR-AUC at each fold to select the best hyperparameter configuration.

To account for the strong class imbalance, all models were trained using class weights inversely proportional to class frequencies, ensuring that the minority class received greater penalization during training.

The best configuration for each model was then evaluated on the validation set, again using PR-AUC as the primary metric.

PR-AUC was preferred over ROC-AUC because it focuses on the positive class, is robust to class imbalance and allows fair model comparison without assuming a fixed decision threshold. Additional evaluation metrics included accuracy, precision, recall, F1-score, F1.5-score and ROC-AUC.

Once the best models were selected based on PR-AUC, classification thresholds were optimized by maximizing

the F1.5-score (Eq. (1)) on the validation set. This metric assigns greater weight to recall than precision, reflecting the clinical priority of capturing all patients at risk of death, even at the cost of an increased number of false positives.

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}, \quad \beta = 1.5 \quad (1)$$

Both hard-voting and soft-voting ensemble strategies were investigated to combine the strengths of multiple classifiers. Hard voting classifies a patient as positive if at least one model predicts mortality:

$$\hat{y}_{\text{hard}}(x) = \begin{cases} 1 & \text{if } \exists m : p_m(x) \geq \tau_m \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Soft voting instead averages the predicted probabilities across models and applies a single optimized threshold τ :

$$\hat{y}_{\text{soft}}(x) = \begin{cases} 1 & \text{if } \frac{1}{M} \sum_{m=1}^M p_m(x) \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where M is the number of models in the ensemble, x is the input feature vector, $p_m(x) \in [0, 1]$ is the predicted mortality probability for sample x produced by model m , and τ_m is the individually tuned threshold for model m .

Finally, the selected ensemble was retrained on the combined training and validation sets, using the hyperparameters and classification threshold identified during the validation phase, before being evaluated on the held-out test set.

3 Results

3.1 Preprocessing Results

The preprocessing phase revealed several common challenges associated with large clinical databases, including missing values, heterogeneous measurement units, duplicated patient records and highly correlated variables.

The mortality outcome was strongly imbalanced, with fewer than 10% of patients classified as deceased. This imbalance motivated the use of evaluation metrics specifically designed for rare-event prediction and the adoption of threshold optimization strategies favoring sensitivity.

Missing value analysis showed great variability across features. Several variables with extremely high percentages of missing observations were removed before model development, while the remaining missing values were handled using MICE.

Correlation analysis identified multiple highly correlated laboratory measurements. To reduce redundancy and improve model stability, only one representative variable from each correlated group was retained. The following variables were removed:

- **Hemoglobin (Hgb)** and **Red Blood Cells (RBC)**: both highly correlated with Hematocrit (Hct), which was retained as the most clinically informative summary of red blood cell concentration.
- **Mean Corpuscular Hemoglobin (MCH)**: highly correlated with Mean Corpuscular Volume (MCV), which was retained.
- **Chloride**: highly correlated with sodium, which was retained as the primary electrolyte marker.

Exploratory analysis also revealed the presence of physiologically implausible values and outliers in several variables. These observations were either corrected when clear data-entry errors were identified or treated as missing values before imputation. The final dataset combined demographic information, diagnosis categories, laboratory measurements, vital signs and intake/output variables into a unified patient-level representation of 1,003 patients and 80 features. Nevertheless, we decided to train and validate our models also on a reduced dataset; indeed, after applying mRMR feature selection, 40 features were retained, approximately half of the original set, resulting in a final dataset of 1,003 patients and 40 features used for model development.

3.2 Model Performance

The predictive performance of four machine learning algorithms was evaluated: Lasso Logistic Regression, Random Forest, Support Vector Machine (SVM) and XGBoost. Models were trained both on the complete feature set and on the subset of 40 features selected through mRMR. Feature selection consistently improved performance across most algorithms, suggesting that removing redundant variables reduced noise and improved generalization. Consequently, the reduced feature set was adopted for subsequent analyses.

Among the individual classifiers, Random Forest and XGBoost achieved the highest PR-AUC values on the validation set (Table 1), while SVM showed the weakest performance and was therefore excluded from the final ensemble construction. Lasso Logistic Regression was retained because of its strong interpretability and competitive performance despite its relative simplicity.

Model	All features	40 features
Random Forest	0.2542	0.2750
XGBoost	0.2654	0.2701
Lasso Logistic Regression	0.2439	0.2628
SVM	0.2253	0.2164

Table 1: PR-AUC on the validation set for each model trained on the complete and reduced feature sets.

3.3 Threshold Optimization and Ensemble Model

Because the default classification threshold of 0.5 is rarely optimal in highly imbalanced datasets, probability thresh-

olds were tuned for each model by maximizing the F1.5-score on the validation set. This procedure increased sensitivity, although at the cost of a moderate reduction in precision. As shown in Table 2, the optimal threshold was different across models, reflecting differences in the calibration of their predicted probabilities. In particular, XGBoost required an extremely low threshold (0.02), suggesting that the model systematically assigned low probability scores to positive cases, likely due to the strong class imbalance, and therefore needed to operate at a very aggressive decision boundary to achieve adequate recall.

Model	Threshold
Lasso Logistic Regression	0.55
Random Forest	0.22
XGBoost	0.02

Table 2: Optimal classification thresholds.

Both hard-voting and soft-voting ensemble approaches were investigated.

Soft voting outperformed hard voting, achieving higher recall and better overall F1.5-scores. Among all tested combinations (Table 3), the ensemble composed of Lasso Logistic Regression and Random Forest achieved the most favorable trade-off between sensitivity and precision on the validation set. This model was therefore selected as the final predictive architecture.

Models	F1.5	Recall	Precision
Lasso	0.3929	0.5500	0.2391
RF	0.3611	0.7500	0.1667
XGB	0.3316	0.5000	0.1887
Lasso+RF	0.3874	0.9000	0.1698
Lasso+XGB	0.3724	0.5500	0.2157
RF+XGB	0.3482	0.7500	0.1579
Lasso+RF+XGB	0.3849	0.9000	0.1682

Table 3: Soft voting ensemble results on the validation set.

After retraining, the ensemble outperformed both individual models on the held-out test set, confirming that the performance gains observed during validation generalize to unseen data as shown in the test set confusion matrix (see Figure 2).

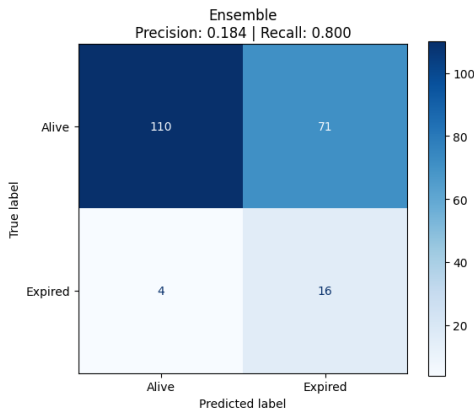


Fig. 2: Confusion matrix ensemble model on test set.

3.4 Model Interpretability

To better understand the clinical factors driving mortality prediction, three interpretability approaches were applied: Lasso coefficients, permutation feature importance and SHAP values. The agreement between multiple interpretability techniques, as shown in Table 4, suggests that the model relies on clinically meaningful signals.

Feature	Lasso	Permutation	SHAP
Age	+0.538	✓	✓
Systolic BP (max)	+0.432	✓	✓
RDW% (max)	+0.341	✓	✓
Heartrate	+0.232	✓	✓
Respiratory rate	+0.155	✓	✓
Endocrine/metabolic disorders	-0.599	✓	✓
Blood/blood organs disorders	-0.351		
Pulse Pressure (PP)	-0.196	✓	✓
Systolic BP (mean)	-0.191		✓
Calcium (mg/dL) (mean)	-0.188		✓

Table 4: Top 10 prognostic features by absolute Lasso weight, with agreement across interpretability methods.

The Lasso coefficients reveal a clinically coherent risk profile:

- **Age** ($w = 0.538$): the strongest predictor, consistent with the known association between advanced age and higher ICU mortality risk.
- **Maximum systolic blood pressure** ($w = 0.432$) and **heart rate** ($w = 0.232$): capture the inability of the cardiovascular system to maintain stable blood flow and pressure.
- **Maximum RDW%** ($w = 0.341$) and **respiratory rate** ($w = 0.155$): reflect systemic inflammation and acute physiological stress, both strongly linked to poor ICU outcomes.
- **Endocrine and metabolic disorders** ($w = -0.599$) and **blood disorders** ($w = -0.351$): the negative sign may seem counterintuitive, but patients with these chronic conditions are often already under close clinical supervision, which may reduce their short-term risk of death.

3.5 Stratified Performance Analysis

A subgroup analysis was conducted to evaluate model performance across different patient demographics and hospital characteristics. Overall, the model showed relatively stable performance across most strata. However, some variability in sensitivity was observed across gender, ethnicity, hospital size and geographic region. Female patients showed higher sensitivity than male patients, while larger hospitals tended to exhibit lower recall than medium-sized institutions. Nevertheless, most subgroup differences should be interpreted cautiously because of the limited number of observations within several categories. Overall, the stratified analysis did not reveal strong evidence of systematic performance failures, although larger

datasets would be required to perform a more robust fairness assessment.

4 Discussion and Conclusion

This study demonstrates the feasibility of predicting in-hospital mortality using only information available during the first 24 hours of ICU admission. The results confirm that early clinical, laboratory and physiological data contain sufficient information to identify patients at increased risk of adverse outcomes.

Among the evaluated algorithms, ensemble methods achieved the best overall performance. In particular, the soft-voting combination of Lasso Logistic Regression and Random Forest provided the most favorable balance between predictive accuracy and sensitivity.

The proposed model achieved particularly strong recall, successfully identifying the large majority of truly at-risk patients. This came at the cost of a higher number of false positives, which is a direct consequence of the aggressive threshold tuning driven by the clinical objective of minimizing missed deaths. Indeed, our model correctly classified as dead 16 out of 20 patients, marking as false negatives only 4 of them; however, the number of false positives was significantly higher (71 out of 181). Importantly, this trade-off is adjustable: if a higher precision is desired, for instance, to reduce unnecessary clinical interventions, the classification threshold can be increased without retraining the model, offering flexibility for deployment in different clinical contexts.

Interpretability analyses performed through Lasso coefficients, permutation importance and SHAP values consistently identified a small set of highly informative predictors, which increases confidence in the clinical relevance of these features and suggests that the model captures physiologically meaningful patterns rather than random associations.

The proposed deployment flow, illustrated in Figure 3, shows how the model could be integrated into clinical scenarios: physicians would receive a mortality risk score derived from a compact set of routinely collected variables within the first 24 hours of admission.

A stratified evaluation across demographic and hospital-level subgroups revealed generally stable performance, with differences across groups that should be interpreted carefully due to their numerosity.

Nevertheless, this analysis provides an important first assessment of potential performance disparities and highlights the importance of fairness evaluation in clinical machine learning applications.

Regarding regulatory and ethical compliance, any clinical deployment of this model would require adherence to the EU AI Act, under which clinical decision support systems are classified as high-risk AI, implying mandatory conformity assessment, transparency, human oversight and post-market monitoring. Patient data must comply with GDPR regulations and model predictions

must always be reviewed by qualified medical personnel, as the model is intended as a decision-support tool and not a substitute for clinical judgment.

Several limitations should be acknowledged. The strong class imbalance, the presence of missing values requiring imputation and the lack of external validation represent the main constraints on the generalizability of the proposed approach. The model can nonetheless be continuously improved as new data becomes available: a batch update strategy (i.e. periodically retraining the full pipeline on accumulated data) is recommended given that Lasso and Random Forest do not support incremental learning. Alternatively, replacing them with online-learning models (e.g. Stochastic Gradient Descent Classifier) would enable updates without full retraining, at the cost of changing the model architecture.

Future work could explore temporal deep learning architectures, larger multicenter datasets and advanced ensemble strategies capable of modeling longitudinal patient trajectories. External validation and calibration analyses would also be valuable for assessing model generalizability and clinical utility. In conclusion, the proposed framework successfully combines rigorous preprocessing, feature selection, machine learning modeling and interpretability analysis to predict ICU mortality.

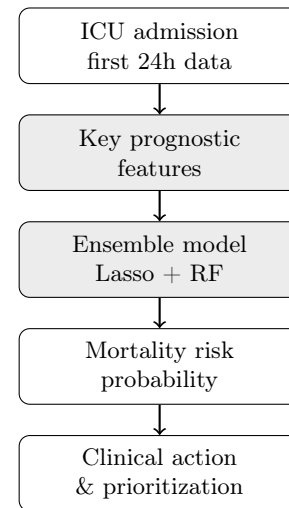


Fig. 3: Proposed clinical deployment flow.

5 References

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